COURSE OUTLINE



DATA9101

Business Intelligence Essentials

Fall 2020 - Current

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DATA9101 Business Intelligence Essentials

COURSE DESCRIPTION

This foundational course introduces essential concepts and skills in business intelligence analysis to prepare you for success in the IBM Business Intelligence Analyst badge course. The focus is on gathering, organizing, analyzing, and visualizing data to improve business decision-making.

REQUISITES	None
EQUIVALENTS	None
CREDITS	0
HOURS	24
ELIGIBLE FOR PLAR	No

COURSE LEARNING OUTCOMES

Bow Valley College is committed to ensuring our graduates can demonstrate their abilities in key areas that will make them effective citizens and encourage their development as lifelong learners. In addition to the discipline-specific skills that learners acquire in their programs, the College has identified ten learning outcomes.

College-Wide Outcomes:

1. Communication
2. Thinking Skills
3. Numeracy and Financial Literacy
4. Working with Others
5. Digital Literacy
6. Positive Attitudes and Behaviours
7. Continuous Learning
8. Health and Wellness Awareness
9. Citizenship and Intercultural Competence
10. Environmental Sustainability

#	COURSE LEARNING OUTCOME(S)	COLLEGE WIDE OUTCOMES SUPPORTED
1	Explain types of data analytics as well as the tools and techniques available for these analytics	Thinking Skills (2), Digital Literacy (5) Numeracy and Financial (3) Communication (1)
2	Choose appropriate charts or graphs for different tasks including, data set comparison, data relationships, data distributions, and composition of data	Thinking Skills (2), Digital Literacy (5) Numeracy and Financial (3) Communication (1)
3	Utilize data visualizations to tell stories and show insights	Thinking Skills (2), Digital Literacy (5) Numeracy and Financial (3) Communication (1)
4	Describe the fundamentals and workflows for prescriptive analytics	Thinking Skills (2), Digital Literacy (5) Numeracy and Financial (3) Communication (1)

5	Perform basic data analytics using real world dataset and draw conclusions using calculated statistics	Thinking Skills (2), Digital Literacy (5) Numeracy and Financial (3) Communication (1)

COURSE MODULES AND SCHEDULE

**Course schedule subject to change, depending on delivery mode and term of study. For exact dates, please consult the Course Offering Information in Brightspace.*

WEEK/HOURS	MODULES
Week 1	Introduction: Module 1 – What is data analytics • Descriptive analytics • Exploratory analytics • Explanatory analytics
Week 2	Module 1 – What is data analytics • Descriptive analytics • Exploratory analytics • Explanatory analytics, Quiz 1
Week 3	Module 2 – Decision making fundamentals, phases, and concepts. • Foundations of decision making. • Simon's four phases for decision making. • Basic concepts of a data warehouse • Data warehouse architecture • Processes used in developing and managing data warehouses
Week 4	Module 2 – Decision making fundamentals, phases, and concepts. • Foundations of decision making. • Simon's four phases for decision making. • Basic concepts of a data warehouse • Data warehouse architecture • Processes used in developing and managing data warehouses
Week 5	Module 3 –Business reporting fundamentals, definitions, and concepts • Business reporting definitions and concepts • Data and information visualization • Importance of data and information visualization • Different visualization techniques • Value of visual analytics for BI/BA • Capabilities and limitation of dashboards, Quiz 2
Week 6	Module 4 – Data visualization fundamentals • Rules of effective data analytics • Comparing data sets • Data relationships • Distribution of data • Composition of data • Geographic data
Week 7	Module 4 – Data visualization fundamentals • Rules of effective data analytics • Comparing data sets • Data relationships • Distribution of data • Composition of data • Geographic data
Week 8	Module 5 – Data visualization - storytelling • Getting the idea of the story • Story structure • Story mechanism • Final touches
Week 9	Module 6 – Prescriptive analytics • Fundamentals of prescriptive analytics • Exploring the building blocks of prescriptive analytics • Establishing prescriptive analytics • Exploring trigger event • Formulating trigger event • Formulating business hypotheses • Refining business analytics • Reaching definitive conclusions
Week 10	Module 6 – Prescriptive analytics • Fundamentals of prescriptive analytics • Exploring the building blocks of prescriptive analytics • Establishing prescriptive analytics • Exploring trigger event • Formulating trigger event • Formulating business hypotheses • Refining business analytics • Reaching definitive conclusions, Quiz 3
Week 11	Module 7 – Data analytics in business world • Predictive vs prescriptive analytics • Asking the right questions • Unlocking the data within • Understand averages • Sampling • Forecasting • Correlation vs. causation
Week 12	Module 7 – Data analytics in business world • Predictive vs prescriptive analytics • Asking the right questions • Unlocking the data within • Understand averages • Sampling • Forecasting • Correlation vs. causation
Week 13	Project 1
Week 14	Study Week
Week 15	Study Week & Final Exam

ASSESSMENT

COURSE LEARNING OUTCOME(S)	ASSESSMENT	WEIGHT
1	Quiz 1 – What is data analytics	15%
2, 3, 4	Quiz 2 – Data visualization	15%
5, 6	Quiz 3 – Prescriptive analytics	15%
7	Project 1 – Data analysis with real world data	15%
1, 2, 3, 4, 5, 6, 7	Final Exam – Comprehensive module 1 to 5	40%

Important: For details on each assignment and exam, please see the Course Offering Information.

PERFORMANCE STANDARDS

A grade of P is required to pass this course. Learners may be required to pass this course in order to progress in the program or to meet specific program completion requirements.

Please consult with the program area or the Bow Valley College website for further details.

GRADING SCHEME

Symbol	Description	Value in GPA
P	Pass	N/A
F	Fail	N/A

REQUIRED LEARNING RESOURCES

Additional learning resources may be found in the Course Offering Information or in Brightspace.

ADDITIONAL INFORMATION

Additional information may be found in the Course Offering Information or in Brightspace.

ACADEMIC ACCOMMODATIONS

Learners with a disability (learning, physical, and/or mental health) may qualify for academic and exam accommodations. For more information, or to apply for accommodations, learners should make an appointment with Accessibility Services in the Learner Success Services (LSS) Department. Accessibility Services can also assist learners who may be struggling with learning but do not have a formal diagnosis. To make an appointment visit LSS on the first floor of the south campus or call 403-410-1440. It is the learner's responsibility to contact Accessibility Services and request academic accommodations. For more information, please visit our website at http://www.bowvalleycollege.ca/accessibility.

INSTITUTIONAL POLICIES

Bow Valley College is committed to the highest standards of academic integrity and honesty. Learners are urged to become familiar with and uphold the following policies: Academic Honesty (500-1-7), Learner Code of Conduct, Procedures and Guidelines (500-1-1), Learner Appeals (500-1-12), Attendance (500-1-10), Grading (500-1-6), Academic Continuance and Graduation (500-1-5), and Electronic Communications (300-2-13). Audio or video recording of lectures, labs, seminars, or any other teaching and learning environment by learners is allowed only with consent of the instructor as part of an approved accommodation plan. Recorded material is to be used solely for personal study and is not being used or distributed without prior written consent from the instructor.

Turnitin:

Students may be required to submit their course work to Turnitin, a third-party service provider engaged by BVC. Turnitin identifies plagiarism by checking databases of electronic books and articles, archived webpages, and previously submitted student papers. Students acknowledge that any course work or essays submitted to Turnitin will be included as source documents in the Turnitin.com reference database, where it will be used solely to detect plagiarism. The terms that apply to a student's use of Turnitin are described on Turnitin.com.

Online Exam Proctoring:

Examinations for this course may require proctoring through an online proctoring service. Online proctoring enables online exam taking within a controlled and monitored environment, thereby enhancing academic integrity. Online proctoring may occur through a variety of methods, including but not limited to:

a. live online proctoring where a remote invigilator authenticates identity and observes completion of an exam using specialized software and recordings;
b. automated proctoring where the exam session is recorded and AI (artificial intelligence) analyzed;
c. browser lockdown that limits access to other applications, websites, copying, printing, screen capture and other functions; or
d. a combination of both live/automated proctoring and browser lockdown.

Course instructors will review recordings, analyses, and data obtained through online proctoring for academic integrity infractions. It is the student's responsibility to meet the technical, software, location, and identity verification requirements necessary to enable online proctoring.

Further details of these policies are available in the Academic Calendar and on the Bow Valley College website, bowvalleycollege.ca.

Learners are encouraged to keep a copy of this course outline for future reference.